

INSTITUTE OF ENGINEERING & TECHNOLOGY, DAVV, INDORE

B.E. First Year (ITA and ITB) Class Test-II

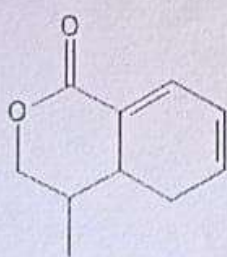
1ACRC2-Chemistry and Environmental Science

Time: 70

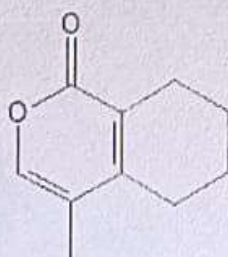
Max Marks: 20

Note: Attempt any four questions. Answer must be brief and to the point. Avoid use of additional Answer Sheet(s).

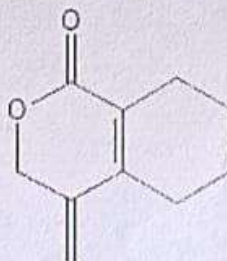
- Q.1** State and derive Beer's-Lambert's law. Calculate percentage reduction in intensity of incident light when 2.5×10^{-5} M aqueous solution of KMnO_4 with molar extinction coefficient $2.0 \times 10^4 \text{ Lmol}^{-1}\text{cm}^{-1}$ is passed through 2.0 cm cell.
- Q.2** What is the basic principle of IR absorption by molecules? Calculate force constant for C=O bond in formaldehyde using Hooke's law approximation if vibrational frequency is 1720 cm^{-1} .
- Q.3** Explain Woodward-Fieser rules in brief. Can you distinguish between the following isomeric compounds by UV spectroscopy?



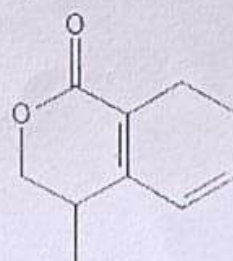
a



b



c



d

- Q.4** Explain important features of IR spectrum of benzoic acid and phenol.
- Q.5** "Delhi, the capital city of India, has been the most polluted capital of the world for the last many years." In light of the above statement, write your views elaborating the nature of pollution and its remedies.
